MIL-T-28681B
11 April 1986
SUPERSEDING
MIL-T-28681A
24 November 1980

MILITARY SPECIFICATION

TIP, DRIVING, STEEL PILE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers cast steel pile point driving tips for protecting the entire end bearing of the steel pile, as a unit, from bending, prevent tipping the flanges from the web, and for frictional resistance of the steel pile.

1.2 Classification.

1.2.1 Types and sizes. In accordance with this specification, driving tips for steel piles shall be of the following types, classes, and sizes as specified (see 6.2).

Type I - Pipe pile tips.
Class 1 - For open-end piles.

Nominal outside diameter in.	Wall thickness in.
10	0.250
12	0.250
14	0.375
16	0.375

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5680

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Class 2 - For closed-end piles.

Nominal outside diameter in.	Wall thickness in.
8	0.250
10	0.250
12	0.250
14	0.375
16	0.375

Type II - Structural HP shape pile tips.

Class 3 - For H-piles.

Nominal size in.	weight lb/ft.
HP 8 x	36
HP 10 x	42
	x 57
HP 12 x	53
	x 63
	x 74
HP 13 x	60
	x 73
	x 87
	x 100
HP 14 x	73
	x 89
	x 102

1.3 Part number systems. Specification part numbers for items described in this specification will be formulated in section 3.8.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-P-116 - Preservation - Methods of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-794 - Parts and Equipment, Procedures for Packaging of.

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(Copies of specifications, standards, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A27 - Mild- to Medium Strength Carbon-Steel Castings for General Application.
A148 - High-Strength Steel Castings for Structural Purposes.
D3951 - Commercial Packaging, Standard Practice for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. Tips for protection of steel piles during installation and permanently under load shall be cast steel meeting the requirements of ASTM A27 or ASTM A148. Bearing between pile and tip shall be square and true with a bearing ledge free of dross and foreign matter.

3.1.1 Design for open-end pipe. For open-end pipe cast-steel cutting shoes may be inside flanged with a 9/16-inch or 5/8-inch ledge for bearing and an outside diameter not more than 1/4 inch over the pipe size. The nearly flush size will assit in maintaining maximum friction along the outside of the pipe. Outside flanged cutting shoes may have an outside diameter not more than 2 inches larger than the pipe size.

3.1.2 Design for closed-end piles. Driving tips for closed-end piles shall be of conical point with an angle of 60 degrees slope to keep soil disturbance to a minimum, and where boulders or sloping rocks are encountered, shall permit development of proper restraint to the entire periphery of the

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pile for frictional resistance and beneath the tip for end bearing. Conical points may be inside or outside-flanged. An outside flange may be tapered for a driving fit to a configuration proven by at least two years of satisfactory service. Inside flanged points shall have a bearing ledge at least 1/2-inch wide for welded attachment.

3.1.3 Design for structural shape piles. Driving tips for structural shapes piles shall have backup on the inner edges and corners of the H-shape encasing the web and backing up the flanges to assure alignment and prevent separation. When specified (see 6.2), the soil surface of the driving tips for structural shape piles shall have seven protruding "teeth" cast integrally with the tip to give better penetration through obstructions, secure a firm bedding and to provide better resistance to movement on a sloping surface.

3.2 First article. When specified in the contract or purchase order (see 6.2), a driving tip shall be subjected to first article inspection (see 4.2.1 and 6.4).

3.3 Standard commercial product. The driving tip shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the driving tip being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.4.1 Steel casting. Unless otherwise specificied (see 6.2), the material shall be steel produced to meet the requirements of ASTM A27 or ASTM A148. Grades of steel shall be specified in 6.2.

3.5 Finish. Unless otherwise specified (see 6.2), the driving tips shall be furnished as cast, except that non-functional sharp points and burrs shall be removed.

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3.6 Identification marking. Identification shall be permanently and legibly marked directly on the driving tip or on a corrosion-resisting metal plate securely attached to the driving tip at the source of manufacturer. Identification shall include the manufacturer's model and serial number, name and trademark to be readily identifiable to the manufacturer.

3.7 Workmanship.

3.7.1 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

3.7.2 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.8 Definitive specification part number. The specification part number is a definitive part number which will be formulated to identify each item covered by this specification.

3.8.1 types shall be as in table I.

TABLE I

Type Code	Remarks
A	Type I - Pipe pile tips.
B	Type II - Structural HP shape pile tips.

3.8.2 classes shall be designated as in table II, III, and IV.

TABLE II. Class II open-end pile.

Class Code	Remarks, O/D x wall thickness/in.
C	10 x .250 inch.
D	12 x .250 inch.
E	14 x .375 inch.
F	16 x .375 inch.

TABLE III. Class III closed-end piles.

Class Code	Remarks, O/D x wall thickness/in.
G	8 x .250 inch.
H	10 x .250 inch.
I	12 x .250 inch.
J	14 x .375 inch.
K	16 x .375 inch.

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TABLE IV. Class III, H-piles.

Class Code	Remarks, sign	in in.	x weight in /1/ ft
L	HP 8 x 36	T	HP 13 x 73
M	HP 10 x 42	U	x 87
N	x 57	V	x 100
O	HP 12 x 53	W	HP 14 x 73
P	x 63	X	x 89
Q	x 74	Y	x 102
R	x 84	Z	x 117
S	HP 13 x 60		

6.6.3 Tooth requirements for Type II, Class 3 shall be as designated in Table V.

TABLE V.

Tooth Code	Remarks
0	No requirement for teeth.
1	Requirement for teeth.

The part number will be formulated by selecting from the requirements options available in this specification as follows:

Tip Pile:

Definitive Military Specification

Part Number M28681 - X X X
Military Specification Number_____
Type (see 3.6.1)_____
Class (see 3.6.2)_____
Tooth (see 6.6.8)_____

e.g., M28681-BE1 is for a Type II structural HP shape pile tips, 14 IN OD X .375 wall thickness with protruding teeth.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality

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program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one driving tip when a first article is required (see 3.2 and 6.2). This inspection shall be the examination of 4.4. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4 and the packaging inspection of 4.5. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. All driving tips offered for delivery at one time shall be considered a lot for the purpose of inspection. The inspection level shall be level S-3 and the Acceptable Quality Level (AQL) shall be 2.5 percent defective. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separated from new lots, and shall be clearly identified as reinspected lots.

4.4 Examination. Each driving tip shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirement shall constitute one defect.

4.5 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 5 of this specification.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A. Unpainted driving tips shall be coated with type P-1 preservative in accordance with MIL-P-116.

5.1.2 Level C. Same as level A except that preservative may be any suitable preservative.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A and B. Packing shall be in accordance with MIL-STD-794. Containers shall be selected from table I for the appropriate level.

5.2.2 Level C. The driving tips shall be packed in accordance with ASTM D3951.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The driving tips covered by this specification are intended for use with steel pipes (pipe or structural shape) as protection against end damage.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type and size of driving tip (see 1.2.1).
c. When protruding "teeth" is specified for structural shape driving tips (see 3.1.3).
d. When a first article is required for inspection and approval (see 3.2, 4.2.1, and 6.4).
e. Material, if other than specified (see 3.4.1).
f. Grade of steel required (see 3.4.1).
g. Finish, if other than specified (see 3.5).
h. Level of preservation and level of packing required (see 5.1 and 5.2).

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of DoD Federal Acquisition Regulations (FAR) Supplement 27.410-6, the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of DoD FAR 27.410-6 are not invoked, the data shall be delivered in accordance with the contract requirements.

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6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - ME
Navy - YD
Air Force - 99

Review:

Air Force - 84

Preparing Activity:

Navy - YD

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-28681B	TIP, DRIVING, STEEL PILE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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